DESIGN CODE OF BOOTH MULTIPLIER (VHDL)

src/booth_mul.vhd [VHDL]

```vhdl
----------------------------------------------------------------------------------
-- Company:
-- Engineer:
--
-- Create Date: 09/10/2026 04:26:32 PM
-- Design Name:
-- Module Name: booth_mul - Behavioral
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
use IEEE.NUMERIC_STD.ALL;
--library UNISIM;
--use UNISIM.VComponents.all;

entity booth_mul is
    Port ( a : in signed (3 downto 0);
           b : in signed (3 downto 0);
           p : out signed (7 downto 0));
end booth_mul;

architecture Behavioral of booth_mul is
begin

    process (a , b)
        variable q_pair : std_logic_vector(1 downto 0);
        variable acc : signed (3 downto 0);
        variable q : signed (3 downto 0);
        variable total : std_logic_vector (8 downto 0);

        variable w : integer := 3;
```

```
47
48      begin
49          q_pair := "00";
50          acc := "0000";
51          q := b;
52          total := std_logic_vector(acc) &  std_logic_vector(q) &  q_pair(0);
53
54          for i in 0 to w loop
55
56              q_pair := total(1 downto 0);
57
58              case(q_pair) is
59                  when "01" => acc := acc + a;
60                  when "10" => acc := acc -a ;
61                  when others => null ;
62              end case;
63
64              total := std_logic_vector(acc) &  std_logic_vector(q) &  q_pair(0);
65              total := total(8) & total(8 downto 1);
66              acc := signed(total(8 downto 5));
67              q := signed(total(4 downto 1));
68
69          end loop;
70
71          p <= signed(total(8 downto 1));
72
73      end process;
74
75  end Behavioral;
```

TESTBENCH CODE OF BOOTH MULTIPLIER (VHDL)

src/booth_mul_tb.vhd [VHDL]

```
1  ----------------------------------------------------------------------------------
   --
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date: 09/10/2026 04:26:32 PM
6  -- Design Name:
7  -- Module Name: booth_mul_tb - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
```

```
--
----------------------------------------------------------------------------------
LIBRARY IEEE;
USE IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
USE IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

ENTITY booth_mul_tb IS
    --  Port ( );
END booth_mul_tb;

ARCHITECTURE testing OF booth_mul_tb IS
    COMPONENT booth_mul IS
        PORT (
            a : IN signed (3 DOWNTO 0);
            b : IN signed (3 DOWNTO 0);
            p : OUT signed (7 DOWNTO 0));
    END COMPONENT;
    SIGNAL a : signed(3 DOWNTO 0);
    SIGNAL b : signed(3 DOWNTO 0);
    SIGNAL p : signed(7 DOWNTO 0);

BEGIN
    booth_multiplier_test : booth_mul PORT MAP(
        a => a,
        b => b,
        p => p
    );
    PROCESS

    BEGIN
        a <= "0101";
        b <= "1010";

        WAIT FOR 10ns;

        a <= "0011";
        b <= "1100";

        WAIT FOR 10ns;

        a <= "1010";
        b <= "1100";

```

```
        WAIT FOR 10ns;

        a <= "0001";
        b <= "1000";

        WAIT FOR 10ns;

        a <= "1110";
        b <= "1011";

        WAIT FOR 10ns;

        a <= "1111";
        b <= "1111";

        WAIT FOR 10ns;

        a <= "0000";
        b <= "0000";

        WAIT FOR 10ns;

        a <= "1010";
        b <= "1111";

        WAIT FOR 10ns;

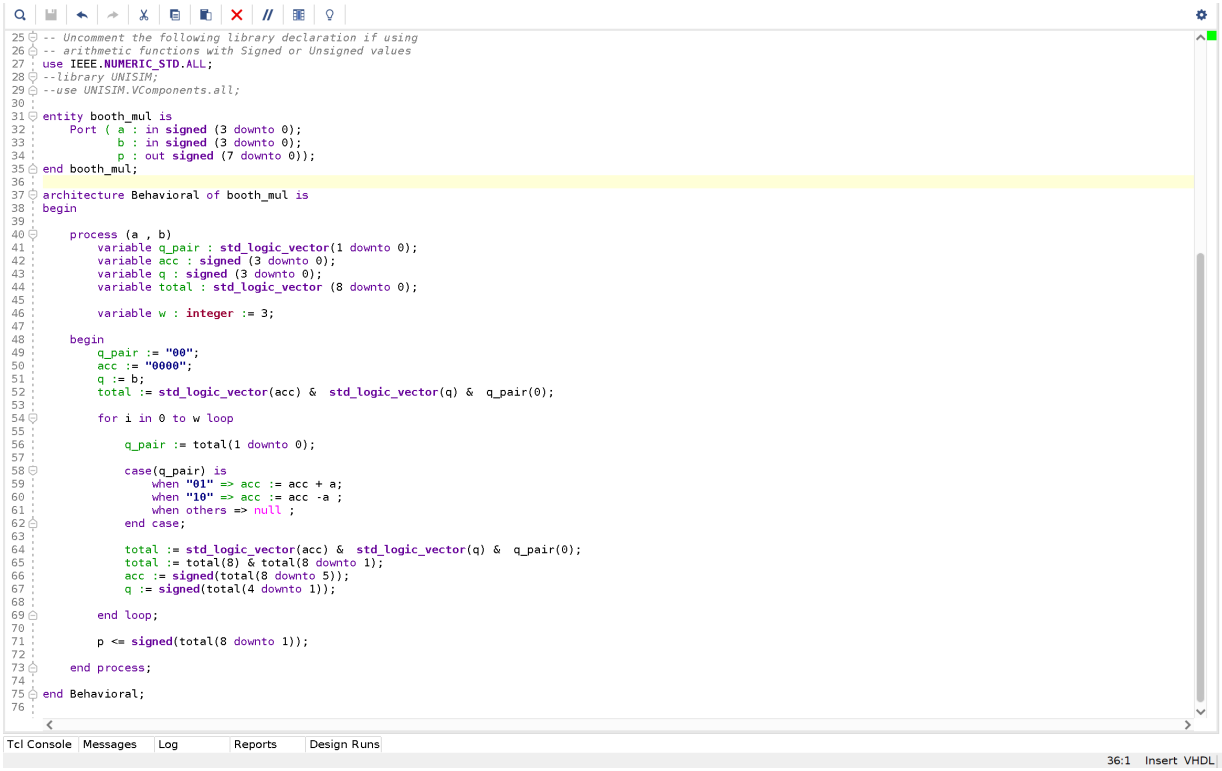
        a <= "0001";
        b <= "1111";

        WAIT FOR 10ns;

        a <= "1010";
        b <= "1111";
        WAIT;

    END PROCESS;
END testing;
```

SCREENSHOT OF CODE AND TESTBENCH



```vhdl
-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
use IEEE.NUMERIC_STD.ALL;
--library UNISIM;
--use UNISIM.VComponents.all;

entity booth_mul is
    Port ( a : in signed (3 downto 0);
           b : in signed (3 downto 0);
           p : out signed (7 downto 0));
end booth_mul;

architecture Behavioral of booth_mul is
begin

    process (a , b)
        variable q_pair : std_logic_vector(1 downto 0);
        variable acc : signed (3 downto 0);
        variable q : signed (3 downto 0);
        variable total : std_logic_vector (8 downto 0);

        variable w : integer := 3;

    begin
        q_pair := "00";
        acc := "0000";
        q := b;
        total := std_logic_vector(acc) &  std_logic_vector(q) &  q_pair(0);

        for i in 0 to w loop

            q_pair := total(1 downto 0);

            case(q_pair) is
                when "01" => acc := acc + a;
                when "10" => acc := acc -a ;
                when others => null ;
            end case;

            total := std_logic_vector(acc) &  std_logic_vector(q) &  q_pair(0);
            total := total(8) & total(8 downto 1);
            acc := signed(total(8 downto 5));
            q := signed(total(4 downto 1));

        end loop;

        p <= signed(total(8 downto 1));

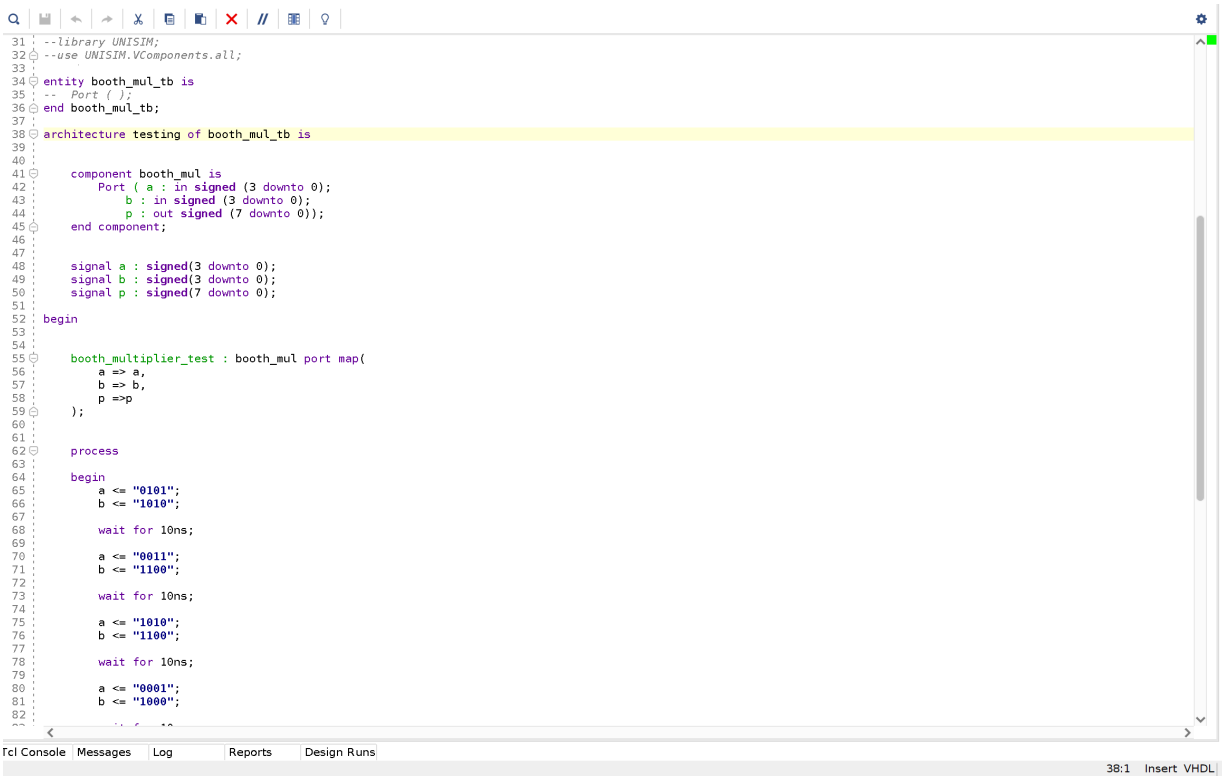
    end process;

end Behavioral;
```

Tcl Console | Messages | Log | Reports | Design Runs

36:1 Insert VHDL

Figure 1: Booth Multiplier VHDL Design Code



```vhdl
--library UNISIM;
--use UNISIM.VComponents.all;

entity booth_mul_tb is
--  Port ( );
end booth_mul_tb;

architecture testing of booth_mul_tb is


    component booth_mul is
        Port ( a : in signed (3 downto 0);
            b : in signed (3 downto 0);
            p : out signed (7 downto 0));
    end component;


    signal a : signed(3 downto 0);
    signal b : signed(3 downto 0);
    signal p : signed(7 downto 0);

begin


    booth_multiplier_test : booth_mul port map(
        a => a,
        b => b,
        p =>p
    );


    process

    begin
        a <= "0101";
        b <= "1010";

        wait for 10ns;

        a <= "0011";
        b <= "1100";

        wait for 10ns;

        a <= "1010";
        b <= "1100";

        wait for 10ns;

        a <= "0001";
        b <= "1000";
```

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38:1 Insert VHDL

Figure 2: Booth Multiplier VHDL Testbench Code

SIMULATION WAVEFORM

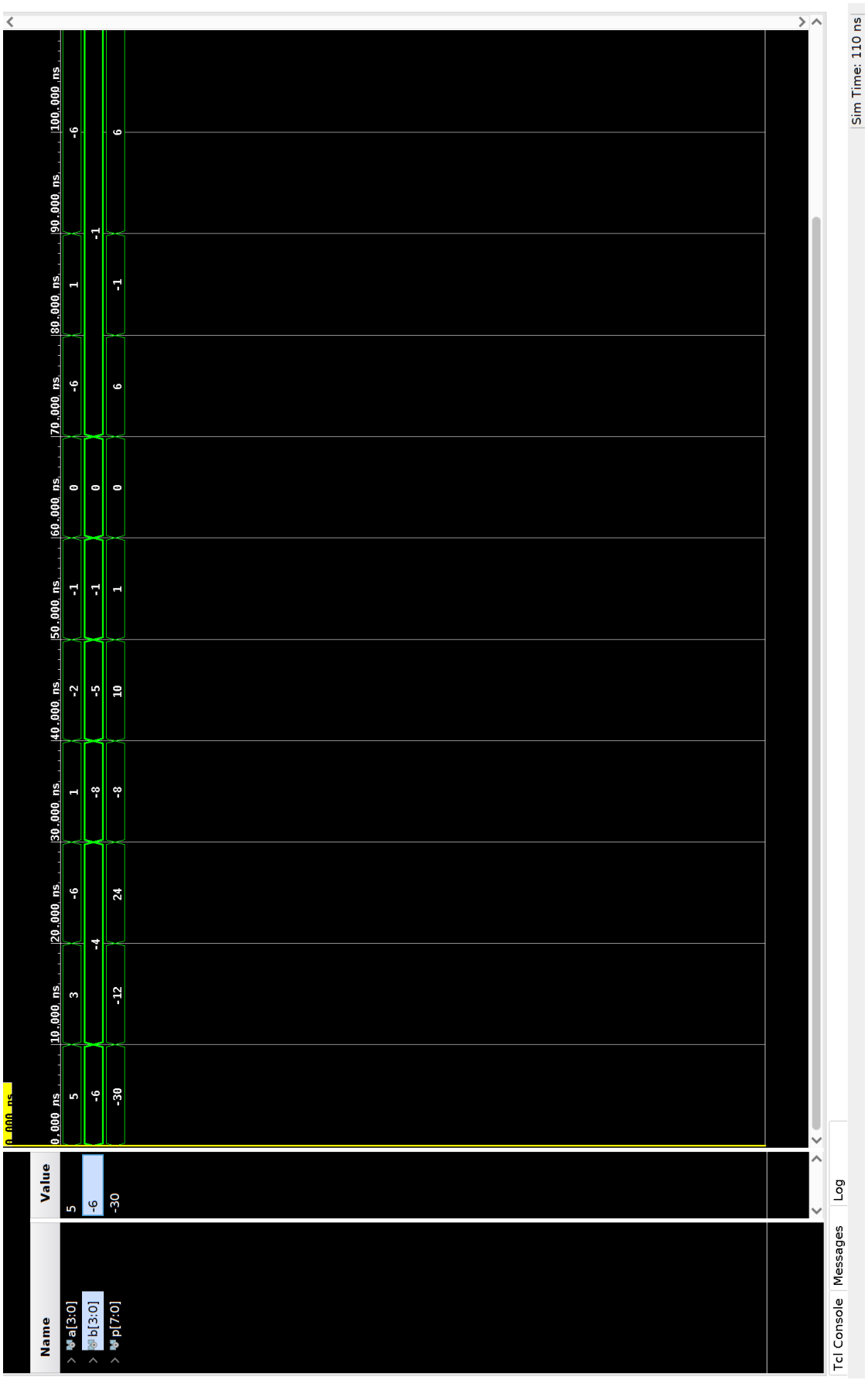


Figure 3: Booth Multiplier Simulation Waveform